Isotope Identification via Gamma Ray Spectroscopy

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November 18, 2025

1 Introduction and Background

During radioactive decay, unstable isotopes of elements decompose into isotopes that are more energetically favorable (i.e., isotopes with lower energy levels) [2]. During a decay process, isotopes randomly emit photons with varying energy levels, which are detectable by scintillation detectors [2, 5]. The frequency distribution of photon energies allows physicists to identify the isotopes that are decaying; this spectrum is a characteristic property of the isotope [2, 5]. This experiment aims to use this fact to identify an unknown isotope by identifying the emission energies corresponding to the isotope's most frequent gamma ray emissions.

In a scintillation detector, incoming gamma rays interact with a crystal like NaI(Tl) via Compton scattering, full absorption, or pair production [2,5]. During full absorption, a gamma ray is absorbed by an electron in the crystal. This electron travels through the crystal, causing secondary excitations of other electrons [2, 9]. The relaxations of these secondary electrons produce scintillation photons, with an aggregate energy equal to the energy of the recoil electron and thus of the original gamma ray [1, 5, 9].

In Compton scattering, a gamma ray photon interacts with a loosely bound electron in the potential well of an atom in the crystal, such as a valence electron, or with similar electrons in materials surrounding the primary detector crystal in the scintillation detector. The photon is redirected at an angle θ relative to its original direction of motion, imparting only *some* of its kinetic energy to the electron (as opposed to all of it, as in full absorption) [2]. If interacting with the crystal, this recoil electron again transfers energy to other electrons, causing secondary photoemissions with aggregate energy equal to the energy of the recoil electron [3, 6]. In contrast, backscattering occurs when the photon interacts with surrounding material with scattering angles close to $\theta = \pi$. In this case, a lower-energy photon is emitted by the material and is absorbed by the crystal, causing a lower-energy recoil electron to be freed [6].

In either case, a photocathode in the scintillation detector converts the secondary photoemissions produced by a recoil electron into a voltage signal representing their total energy—thus recovering the energy of the recoil electron [1, 5].

The resultant energy of the photon involved in the scattering process is E', given by [1, 2, 9]:

$$E' = \frac{E_\gamma}{1 + \frac{E_\gamma}{m_e c^2}(1 - \cos\theta)} \tag{1}$$

where m_e is the mass of the electron, c is the speed of light, and E_γ is the energy of the original gamma ray. Note that full absorption produces the primary, higher-energy photopeak of an isotope's gamma ray spectrum (with energy E_γ), whereas Compton scattering produces photoemissions caused by electrons that only partially inherit the energy of the original gamma ray (with energy $E_\gamma - E'$) [2, 9].

2 Procedure

A scintillation detector paired with a multi-channel analyzer (MCA) is used to obtain a frequency distribution of photon energies, in an arrangement described in Figure 1. Before an unknown element is identified, calibration isotopes with known emission peaks—namely, Co-60, Ba-133, and Am-241—are first used to obtain a mapping from the MCA's bin numbers to the bins' associated photon energies. Each calibration isotope is placed in the detector one by one. Channel numbers corresponding to emission peaks, along with full-width at half-maximum (FWHM) channel number bounds, are recorded, yielding the data in Table 1. A linear regression model is fit to energy vs. channel number data points, yielding a function from channel number to photon energy used in later steps. Note that adjustment of the scintillation detector's input voltage, preamplifier gain, and amplifier gain may be necessary to place the emission spectrum within the MCA's range; these parameters should not be changed during and after calibration.

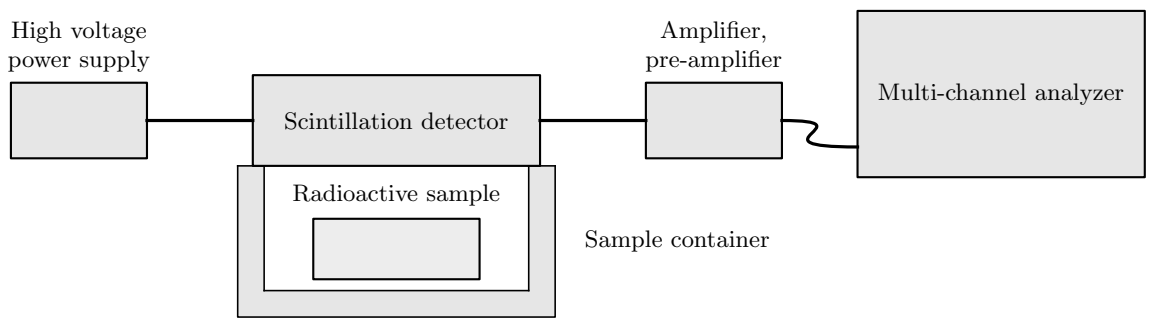

Figure 1: A schematic of the experimental apparatus. A radioactive sample is placed in a container. A scintillation detector, powered by a high-voltage power supply, is placed directly above the sample, collecting gamma rays and causing either full absorption or Compton scattering and converting the resultant secondary photoemissions into voltage signals. A pre-amplifier and amplifier pair adjust the voltage signals to suitable parameters for the MCA: the voltage is set to an expected operation range and voltage pulses are shaped to be well-defined on an oscilloscope.

To verify the correctness of the calibration, a sample of Cs-137 is placed within the detector. The emission spectrum is recorded, along with the peak's centroid channel number and FWHM channel number bounds, producing the data in Figure 4. An entirely unknown isotope is then placed in the detector for identification. Channel numbers of emission peaks and FWHM bounds are again recorded, producing the data points in Table 3.

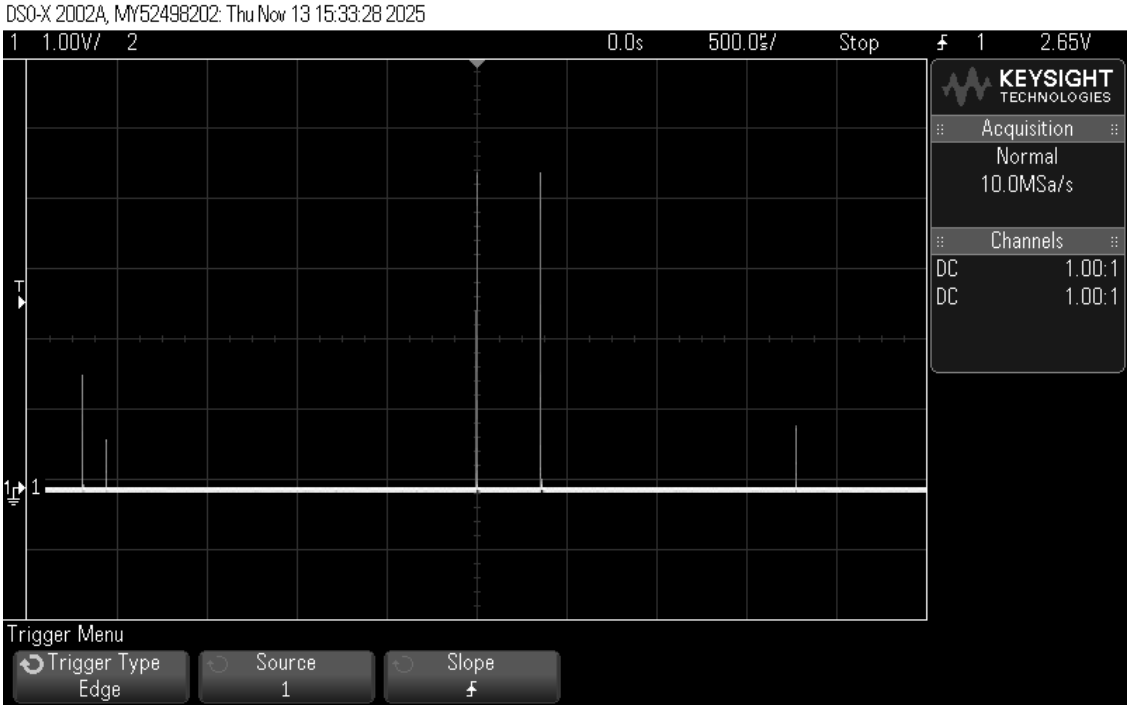

Figure 2: Voltage pulses corresponding to emission events from a Cs-137 sample, recorded on an oscilloscope monitoring the amplifier output. The heights of the pulses are proportional to the aggregate energy of the secondary photoemissions caused by recoil electrons from Compton scattering and full absorption [2]. "Darker" horizontal bands on the oscilloscope trace correspond to low-frequency features of the emission spectrum, such as the space between the Compton edge and the photopeak.

3 Results

The known energies vs. channel number data points used to calibrate the detector and MCA are presented in Table 1, while the linear model providing the conversion from channel numbers to photon energies is displayed in Figure 3.

Isotope	Known Energy (keV)	Channel (Index)	FWHM (Index)	Estimated Energy (keV)	Discrepancy
Co-60	1173.0	790	42	1170 ± 28	0.11σ
Co-60	1332.5	895	44	1326 ± 29	0.22σ
Ba-133	302.9	212	50	310 ± 30	0.24σ
Ba-133	356.0	250	20	364 ± 13	0.62σ
Am-241	59.6	46	5	59 ± 4	0.15σ

Table 1: Known energies, recorded channel numbers, full-width at half-maximum interval widths, and estimated energies for the calibration isotopes Co-60, Ba-133, and Am-241. Note that Co-60 and Ba-133 have two emission peaks, while Am-241 has one. Discrepancies between estimated and known energies lie within 1σ.

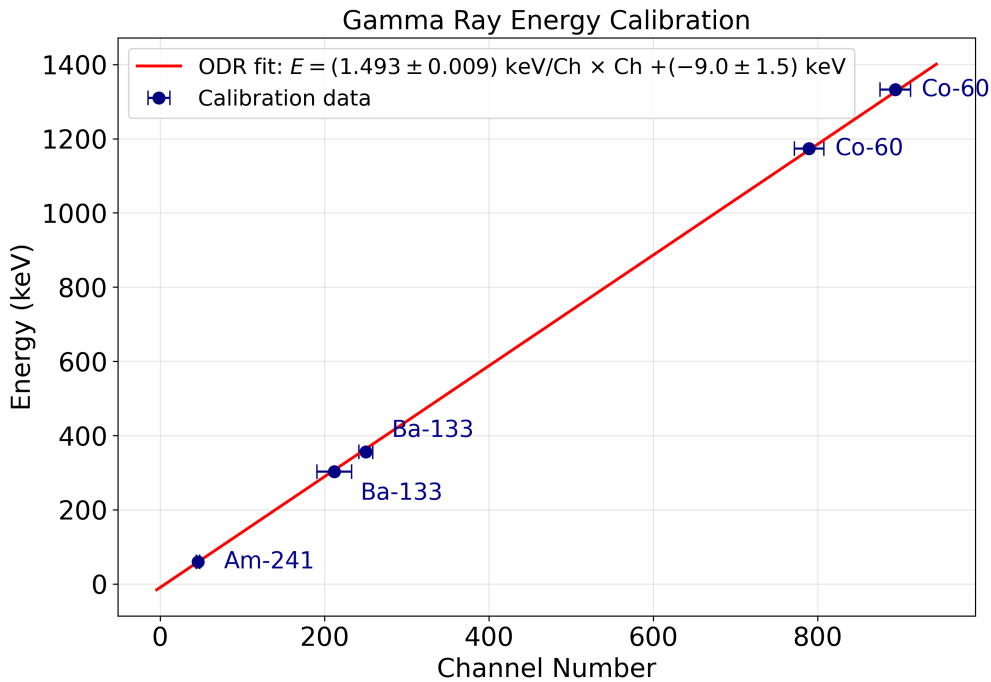

Figure 3: Linear regression model (red) used to convert MCA channel numbers to photon energies, derived from known-energy vs. channel number data points (blue) obtained from the calibration isotopes.

Figure 4 includes a plot of the emission spectrum of Cs-137 with energies obtained from the regression model. Notable features of the spectrum are identified, including the primary photopeak, Compton backscatter peak, Compton plateau, Compton edge, and the X-ray photopeak. Using equation 1 with a known Cs-137 photopeak energy of $E_\gamma = 662$ keV [7], the Compton backscatter peak is calculated to occur at 184.3 keV ($\theta = \pi$), with the Compton edge occurring at 477.4 keV ($\theta = 0$) [2].

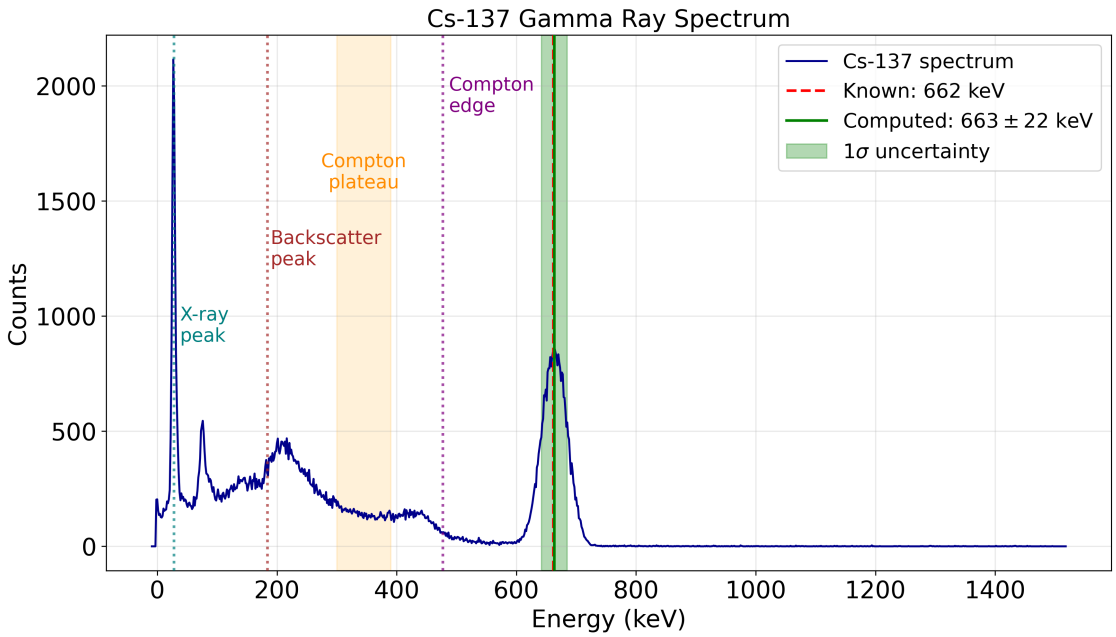

Figure 4: A spectrum of Cs-137 collected from the MCA, with energies derived from channel numbers using the linear regression model from Figure 3. The X-ray peak (light blue), theoretical Compton backscatter peak (red, 184.3 keV), Compton plateau (yellow), and Compton edge (purple, 477.4 keV) are identified. The known full-absorption photopeak (red, 662 keV) and the computed full-absorption photopeak (green, 663 keV ± 22 keV) are also identified.

Table 3 includes the channel numbers and estimated energies of the unknown isotope. The emission peak energies of the unknown isotope are found to agree within 1σ of the known emission peaks of Ra-226 [8], strongly suggesting it is the unknown isotope. The computed peak energies of the unknown isotope in comparison with Ra-226's peak energies are presented in Table 3.

Channel (Index)	Computed Energy (keV)	Ra-226 Candidate Peak (keV)	Discrepancy
131.76	188 ± 5	186	0.35σ
169.86	244 ± 6	242	0.43σ
205.26	297 ± 7	295	0.33σ
245.36	357 ± 8	351	0.75σ
417.46	614 ± 14	609	0.36σ

Table 2: Channel numbers of emission peaks of the unknown isotope, with computed energies and possible Ra-226 emission peaks [8]. Discrepancies between the estimated energies and the known energies of Ra-226 lie within 1σ.

4 Conclusion and Discussion

An MCA paired with a scintillation detector is found to be a reliable tool for producing gamma ray emission spectra of radioactive isotopes. The function from MCA channel numbers resulting from the calibration of such a system is found to reproduce the known energies of the calibration

isotopes within 1σ of accepted values and the emission peak of Cs-137 to within 1σ of the accepted value, suggesting a successful calibration process.

Notable features of the emission spectrum of Cs-137 are identified. The Compton backscatter peak and Compton edge qualitatively occur near the theoretical energies of 184.3 keV and 477.4 keV, with the Compton plateau corresponding to a continuum of possible scattering angles between $\theta = 0$ and $\theta = \pi$. A backscattering peak is formed toward the lower end of the scattering spectrum; this peak results from scattering events corresponding to θ-values between $\theta = \frac{2\pi}{3}$ and $\theta = \pi$, which produce a narrower spread of energies than those corresponding to θ-values between $\theta = 0$ and $\theta = \frac{2\pi}{3}$ [3]. Low-energy x-ray peaks are also observed near 30 keV, corresponding to decay events of Cs-137 into its next decay product, Ba-137, without emitting a typical 662 keV gamma ray. In these cases, an electron occupying a $1s$ state is ejected from the atom, and an electron occupying a higher-energy state within the atom relaxes to fill the $1s$ hole, emitting the lower-energy x-ray photon [4–6].

The unknown isotope investigated is identified to be Ra-226, with all computed emission peaks agreeing within 1σ of the corresponding known emission peaks of Ra-226.

References

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